

Modeling Advance Hodgkin-Huxley Model with Scientific Machine Learning

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1 Introduction

The Hodgkin-Huxley model is a fundamental framework for understanding the electrical characteristics of neurons. It describes the propagation of action potentials using a system of nonlinear differential equations. This project aims to recover the actual reaction dynamics within the model using Scientific Machine Learning techniques.

2 Hodgkin-Huxley Model (Classical)

$$C_m \frac{dV}{dt} = I - g_K n^4 (V - E_K) - g_{Na} m^3 h (V - E_{Na}) - g_L (V - E_L)$$

Where:

- V is the membrane potential,
- C_m is the membrane capacitance,
- I is the input current,
- g_K, g_{Na}, g_L are the conductances of potassium, sodium, and leak channels respectively,
- E_K, E_{Na}, E_L are their respective reversal potentials,
- m, h, n are gating variables that follow first-order kinetics.

3 Advanced HH Model (with Additional Currents)

To better reflect biological complexity, we extend the classical HH model by adding:

- A persistent sodium current: $I_{NaP} = g_{NaP}p(V - E_{Na})$
- A calcium current: $I_{Ca} = g_{Ca}s^2(V - E_{Ca})$

$$C_m \frac{dV}{dt} = I - g_K n^4 (V - E_K) - g_{Na} m^3 h (V - E_{Na}) - g_L (V - E_L) - g_{NaP} p (V - E_{Na}) - g_{Ca} s^2 (V - E_{Ca})$$

Where:

- p : gating variable for persistent sodium channels,
- s : gating variable for calcium channels,
- g_{NaP}, g_{Ca} : maximal conductances,
- E_{Ca} : calcium reversal potential.

4 Gating Dynamics

$$\begin{aligned} \frac{dm}{dt} &= \alpha_m(1 - m) - \beta_m m \\ \frac{dn}{dt} &= \alpha_n(1 - n) - \beta_n n \\ \frac{dh}{dt} &= \alpha_h(1 - h) - \beta_h h \\ \frac{dp}{dt} &= \alpha_p(1 - p) - \beta_p p \\ \frac{ds}{dt} &= \alpha_s(1 - s) - \beta_s s \end{aligned}$$

5 Voltage-dependent Rate Constants (for Classical Gates)

$$\alpha_m = \frac{0.1(V + 40)}{1 - e^{-(V+40)/10}}, \quad \beta_m = 4e^{-(V+65)/18}$$

$$\alpha_h = 0.07e^{-(V+65)/20}, \quad \beta_h = \frac{1}{e^{-(V+35)/10} + 1}$$

$$\alpha_n = \frac{0.01(V + 55)}{1 - e^{-(V+55)/10}}, \quad \beta_n = 0.125e^{-(V+65)/80}$$

(Additional rate constants $\alpha_p, \beta_p, \alpha_s, \beta_s$ to be defined based on specific biological assumptions or learned using UDE.)

6 Objectives

1. Replace the full ODE with a Neural ODE and perform prediction and forecasting.
2. Replace the calcium current term $g_{Ca}s^2(V - E_{Ca})$ with a neural network, forming a Universal Differential Equation (UDE), and recover hidden calcium dynamics from data.
3. Extract the symbolic form of the recovered neural network to interpret the underlying reaction dynamics.

7 Immediate Action Items

1. Implement and run the given ODE in Julia.
2. Generate underlying data, including adding noise to simulate real-life conditions.
3. Implement and run Neural ODEs to get prediction results.
4. Implement and run UDEs to get prediction results.
5. Start a GitHub repository to upload code files and figures.

6. The manuscript for this project is being written in Overleaf and can be accessed [here](#)